



Department of Electronics and Communication Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Innovative Practice Description

- **Unit / Topic:** Unit II / Aperture antennas
- **Course Outcome: CO4** The students will be able to determine the radiation mechanism and design aspects of various antennas.
- **Topic Learning Outcome: TLO9** The students will be able to design Aperture Antennas, Reflector antennas and Microstrip antennas
- **Activity Chosen: Student Team Achievement Divisions (STAD)**
- **Justification:**

Student Team Achievement Divisions (STAD) is a most suited collaborative learning model for a Conceptual course. I have chosen STAD among the four learning model (JIGSAW, STAD, TGT, or TAI) suggested. The different types of Aperture antennas and its design procedure can be studied and learned the applications of aperture antennas. Aperture antennas are classified as Horn antenna, Slot antenna, Reflector antenna and Microstrip patch antenna. STAD is a Collaborative Group activity. We can assign Aperture antennas to the student groups to learn the operations and its design procedures better.

- **Time Allotted for the Activity:** 50 minutes

Details of the Implementation:

Step1: Group Study Time (25 minutes)

- ❖ The time is given for each group to revise/study the Aperture antennas and its design procedures.
- ❖ Each group should study the Horn antenna- Introduction, Types of Horn antenna, Feeding methods, Design procedure and Applications
- ❖ Each group should study the Slot antenna- Introduction, Types of Slot antenna, Feeding methods, Design procedure and Applications
- ❖ The high caliber students take the responsibility of ensuring that every member is well- prepared in the content.

Step 2 – Final Evaluation Quiz – (15 minutes)

- ❖ At the end of the stipulated time, the entire class taken a quiz. The quiz
- ❖ is corrected in class by projecting the correct answers.
- ❖ Individual students' scores have displayed and then group scores can be calculated.

Step 3 - Announcing the Winners (5 minutes)

Top scoring group and Top performer has appreciated and encouraged by the Senior faculty member of the department.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO10	PSO3
CO5	3	2	2	2	3	3	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO8	PO9	PO10	PSO3
	3	3	3	3	2
Justification for correlation	Electromagnetic Fields fundamentals are applied	Group activity which requires inter-personal ethics	Needs individual and team work	Effective communication is required	Students involves design and development of Aperture antennas

- **Images / Screenshot of the practice:**





• **Reflective Critique:**

Benefits:

- ❖ The students involved activity in the activity and discussed with their neighbors.
- ❖ The understanding level of the concept can be identified, while the students are sharing their view.
- ❖ The question was asked in the Assessment test, most of the students answered correctly.

Challenges:

- ❖ Making all the students to share the view was not possible due to time constrain.
- ❖ While randomly choosing the group few members struggle to answer the question.

References:

1. <https://www.scirp.org/journal/paperinformation.aspx?paperid=96719>

Signature
8/9/22

Signature of Faculty Member

Signature
8/9/22
HOD



Department of Electronics and Communication Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Innovative Practice Description

- **Unit / Topic:** Unit IV / Power Divider and Magic Tee
- **Course Outcome: CO4** The students will be able to explain the basic concept of various microwave devices.
- **Topic Learning Outcome: TLO9** The students will be able to describe the basic principles of Microwave Passive components
- **Activity Chosen: Flipped Classroom**
- **Justification:**

Power divider and the magic tee are the passive microwave devices which can be understood easily. Flipped classroom practice was chosen because the topic is easy and interesting for the students to refer by themselves.

- **Time Allotted for the Activity:** 50 minutes

- **Details of the Implementation:**

The students were given with the reference materials to get prepared. The students were asked to make a group among them. The questions were asked from the topic they have prepared. As it is power divider and Magic Tee, most of the questions were related to the ports. As a group, they have discussed and came up with answer for the questions. After that, they were asked to get the class back together to share the individual's group work with everyone. While discussing like this, they can get an in-depth knowledge about the topic and they can clearly express their ideologies related to that topic.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO9	PO10	PSO3
CO5	3	2	3	3	3	2

• **PO / PSO mapped:**

Innovative practice	PO1	PO8	PO9	PO10	PSO3
	3	3	3	3	2
Justification for correlation	Electromagnetic Fields fundamentals are applied	Group activity which requires inter-personal ethics	Needs individual and team work	Effective communication is required	Students involves design and development of microwave passive devices

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ *Feedback of practice from students and other stakeholders:*

1. They can get engaged in class for the entire 50 minutes.
2. The activity acts as a better platform for discussion.
3. It is quite interesting.

❖ *Benefit of the practice:*

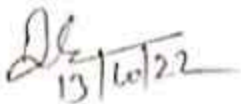
1. They will get exposed to text books and reference materials.
2. Better interaction and communication due to group activity.
3. All the students can effectively take part in the activity and the idea about the topic has been reached more students.

❖ *Challenges faced in implementation:*

1. To organize the activity little more time has been consumed than expected.

References:

1. <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>


13/10/22

Signature of Faculty Member


13/10/22
HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2022 – 2023 (Odd Semester)

Active Learning Activity- Unit-I- Introduction to Microwave Systems and Antennas

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Link budget and link margin	One Minute Paper	25.08.2022 & 5 Minutes
Noise Characterization of a microwave receiver	Think Pair Share	26.08.2022 & 10 Minutes

Link budget and link margin - One Minute Paper





RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

Accredited by NAAC & An ISO 9001:2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Noise Characterization of a microwave receiver - Think Pair Share





RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2022 – 2023 (Odd Semester)

Active Learning Activity- Unit-II- Radiation Mechanisms and Design Aspects

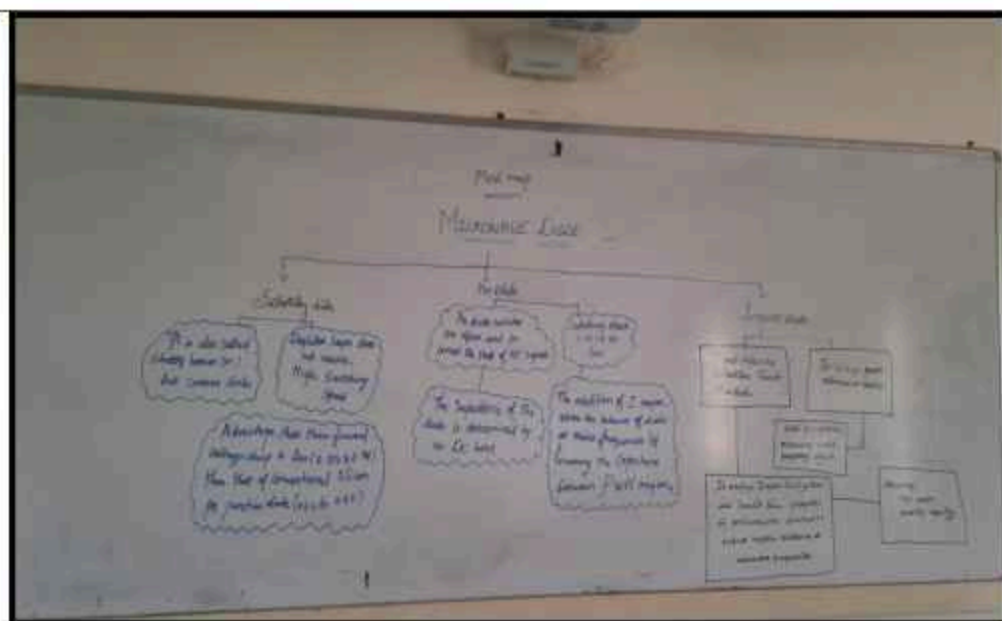
Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Microstrip antennas	Mindmap	06.09.2022 & 10 Minutes

Microstrip antennas - Mindmap





RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2022 – 2023 (Odd Semester)

Active Learning Activity- Unit-III- Antenna Arrays and Applications

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Uniformly spaced arrays with non-uniform excitation amplitudes	Open Book Test	30.09.2022 & 25 Minutes

Uniformly spaced arrays with non-uniform excitation amplitudes - Open Book Test





RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2022 – 2023 (Odd Semester)

Active Learning Activity- Unit-IV- Passive and Active Microwave Devices

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Power Divider, Magic Tee	Hardware Demonstration	10.10.2022 & 10 Minutes

Uniformly spaced arrays with non-uniform excitation amplitudes - Open Book Test



[Signature]
13/10/22

Signature of Faculty Member

[Signature]
13/10/22
HOD